



University of
Inland Norway

ØKA2015 Applied Data Science

Autumn 2026 | 7.5 study points | Lillehammer

Lecture 2

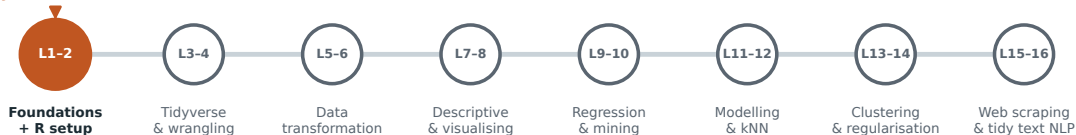
The Data Science Process, Tidy Data, and R

Touseef Hameed | `touseef.hameed@inn.no`

Thursday 27 August 2026 | 12:15–14:00 | Room 1V2062

Where we are in the course

you are here



Last time (Lecture 1, with Fikru)

What data science *is*, why it matters, the three kinds of analytics, big data, and the legal and ethical rules of the game.

Today

How it is actually done, the process from raw file to decision, what “tidy” data means, and getting R running on your own laptop.

Today's plan

① The language of data

variable, observation, value, dataset, database

② The Data Science Process Flow

import, tidy, transform, visualise, model, communicate

③ Tidy data

the three rules that make everything else easy

④ The data science ecosystem

tools, people, processes, infrastructure

⑤ R and RStudio

what they are, and why you need both

⑥ Data Lab, install R and RStudio

By the end you will be able to

- ✓ name the parts of a dataset correctly
- ✓ describe the steps of a data science project
- ✓ tell tidy data from messy data on sight
- ✓ explain how R and RStudio differ
- ✓ run R on your own laptop

You do not need any programming experience. We start from zero.

1. The language of data

First we need to agree on **what to call the parts of a dataset.**

Variable
one column
= one measured attribute
▼

employee_id	name	age	department
001	John Smith	30	Marketing
002	Jane Doe	25	Sales
003	Bob Brown	40	HR

Observation
one row = one unit

Value
one cell = one measurement

Dataset = a table of values

Database = many related datasets (tables) stored together

Five words you will hear in every lecture from now on.

1. The five words

Term	Plain-English meaning	In our employee table
Variable	A characteristic you measure. It fills <i>one column</i> .	age, name, department
Observation	One thing you measured. It fills <i>one row</i> .	Jane Doe's whole record
Value	A single measurement. It fills <i>one cell</i> .	25
Dataset	A whole table of values.	The employee table
Database	Many related tables stored together.	Employees + orders + products

The one sentence to remember

Columns are **variables**. Rows are **observations**. Cells are **values**.

1. Find them in a real spreadsheet

A variable = a column

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
4	1727	Doe	Janice	female	San Diego	CA	Doctor
5	6879	Jordan	Alex	male	Birmingham	AL	Teacher

An observation = a row

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
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Why this matters

Every tool we use this semester assumes your data is shaped *this* way.

If the shape is wrong, nothing works. If it is right, a surprising amount of the analysis turns out to be one line of code.

Quick question

A survey has 200 respondents and asks 12 questions.

How many **observations**? How many **variables**?

1. Find them in a real spreadsheet

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	A	B	C	D	E	F	G
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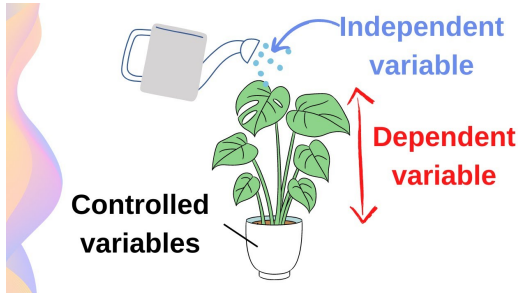
Quick question

A survey has 200 respondents and asks 12 questions.

How many **observations**? How many **variables**?

200 observations, 12 variables.

1. What are variables?



Not every variable plays the same role. Think of an experiment: *does more water make a plant grow taller?*

Independent variable | the one *you* change on purpose. Here: how much water.

Dependent variable | the one you *measure* to see the effect. Here: the plant's height.

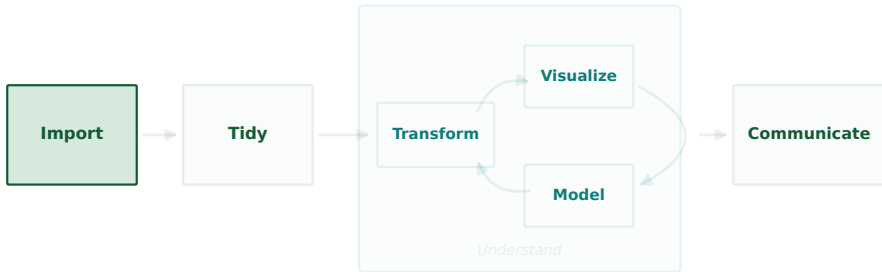
Controlled variables | everything you deliberately hold fixed, so the comparison stays fair. Here: soil quality, sunlight, warmth.

In a dataset

All three are just **columns**. The names describe the *job* each column does in your question, not a different kind of data.

2. The Data Science Process Flow

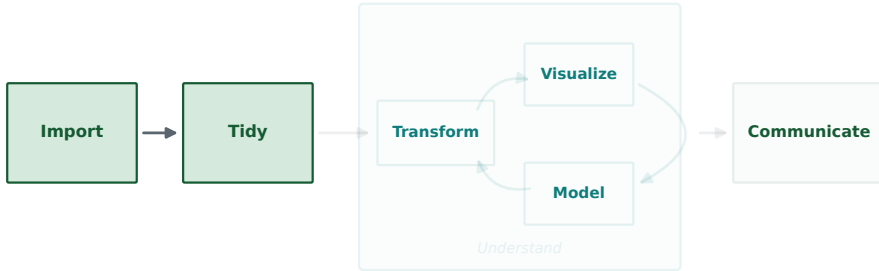
Every data science project, from raw file to final decision.



Adapted from Wickham, Çetinkaya-Rundel & Grolemund, *R for Data Science* (2e), Introduction, <https://r4ds.hadley.nz/intro>

2. The Data Science Process Flow

First: get the data in, and get it into a consistent shape.

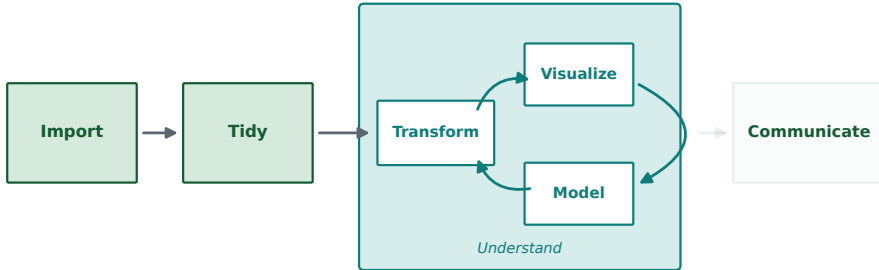


Programming (R) runs underneath every single step

Adapted from Wickham, Çetinkaya-Rundel & Grolemund, *R for Data Science* (2e), Introduction, <https://r4ds.hadley.nz/intro>

2. The Data Science Process Flow

Then: the loop where understanding actually happens.

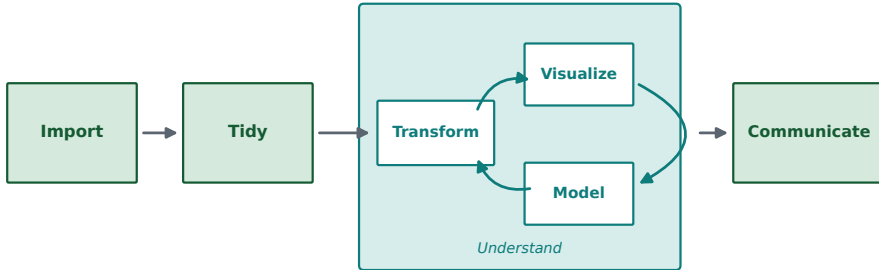


Programming (R) runs underneath every single step

Adapted from Wickham, Çetinkaya-Rundel & Golemund, *R for Data Science* (2e), Introduction, <https://r4ds.hadley.nz/intro>

2. The Data Science Process Flow

Finally: tell someone. If nobody hears it, it may as well not exist.

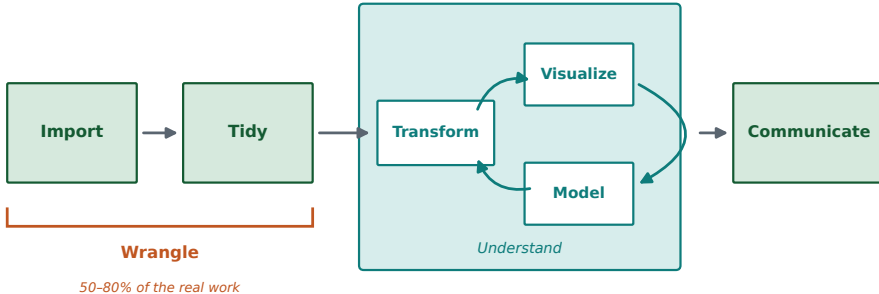


Programming (R) runs underneath every single step

Adapted from Wickham, Çetinkaya-Rundel & Golemund, *R for Data Science* (2e), Introduction, <https://r4ds.hadley.nz/intro>

2. The Data Science Process Flow

And this front half is where your time really goes.



Programming (R) runs underneath every single step

Adapted from Wickham, Çetinkaya-Rundel & Golemund, *R for Data Science* (2e), Introduction, <https://r4ds.hadley.nz/intro>

Step 1: Import

Take data that lives *somewhere else* and load it into R so you can work on it.

Data usually arrives as one of:

A file | a **CSV** (Comma-Separated Values) spreadsheet, an Excel workbook, a text file

A database | many related tables held on a server

An API (Application Programming Interface) | a web address that hands you data automatically

What is a CSV?

A plain text file where a **comma** separates the columns and a **new line** separates the rows. Any program can read it, which is why it is everywhere.

A CSV as text ...

```
day,drink,cups
Mon,Latte,18
Mon,Filter,25
Tue,Latte,22
```

... is this table

day	drink	cups
Mon	Latte	18
Mon	Filter	25
Tue	Latte	22

Same information. One shape is for machines, one is for you.

Step 2: Tidy, the three rules

1

**Each variable
is a column**

2

**Each observation
is a row**

3

**Each value
is a cell**

Get these three right and the rest of the course gets much easier.

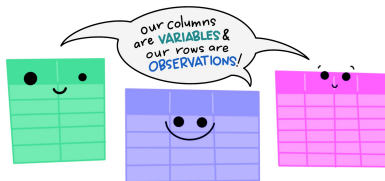
“Tidy datasets are all alike, but every messy dataset is messy in its own way.”

Hadley Wickham

R for Data Science (2e), Ch. 5 *Data Tidying*, <https://r4ds.hadley.nz/data-tidy>

Step 2: Tidy, what messy looks like

The standard structure of tidy data means that "tidy datasets are all alike..."



"...but every messy dataset is messy in its own way."

—HADLEY WICKHAM

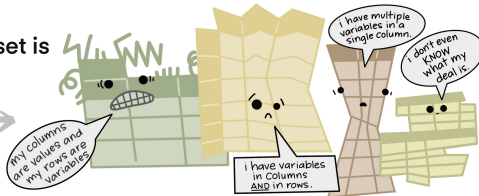


Illustration by Allison Horst (CC BY 4.0).

Step 2: Tidy, our coffee shop

Cups sold at a Lillehammer coffee bar. Same numbers, two shapes.

MESSY (wide)

day	Latte	Filter	Espresso
Mon	18	25	12
Tue	22	19	15
Wed	27	21	11

Column names are drink names.

Those are **values**, not variables.

3 rows, but 9 observations hiding inside them.

tidy it



TIDY (long)

day	drink	cups
Mon	Latte	18
Mon	Filter	25
Mon	Espresso	12
Tue	Latte	22
Tue	Filter	19
Tue	Espresso	15
Wed	Latte	27
Wed	Filter	21
Wed	Espresso	11

9 rows = 9 observations. 3 real variables.

Step 2: Tidy, our coffee shop

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Wed	Filter	21
Wed	Espresso	11

9 rows = 9 observations. 3 real variables.

There is nothing *wrong* with the messy table, a person reads it fine. Software cannot work with it, though, because **Latte** is a value pretending to be a variable.

Steps 3 to 5: Transform, Visualise, Model

Transform means reshaping the data to answer your question:

- keep only the rows you care about
- build new variables out of old ones
- boil many rows down to a summary

(all sales in August)
(revenue = price $\times$ cups)
(average cups per day)

Visualisation

Shows you WHAT is happening.

- + Spots trends and outliers fast
- + Easy to communicate
- Will not tell you WHY
- Does not scale to many variables

Models

Tell you WHY, and what comes NEXT.

- + Handles huge data quickly
- + Quantifies the relationship
- Rests on assumptions
- Blind to patterns never seen before

In practice you go round this loop many times.

Steps 4–5: what a visualisation can look like



Bar charts, scatter plots, line charts, box plots, pie charts, histograms. Each one answers a **different kind of question**.

We build these ourselves with `ggplot2` later in the course.

Step 6: Communicate

The step people forget

A brilliant model that nobody understands or trusts will not change a single decision.

A finished analysis answers, for a non-technical reader:

- What question did you ask?
- What did you find?
- How confident are you?
- **What should we do differently on Monday?**

This is assessed in your assignments, not just the code.

And underneath it all: programming

Every one of these steps is carried out by writing **code**.

Code is what makes an analysis *reproducible*, you or anyone else, can re-run it next year and get exactly the same answer.

That is why we spend the second half of today installing R.

Wrangling: where your time actually goes

Data wrangling

Tidy + Transform

Getting data out of the shape it arrived in and into the shape you need.

It is called wrangling because it genuinely feels like a fight.

50–80%

of a data scientist's time

Set your expectations now

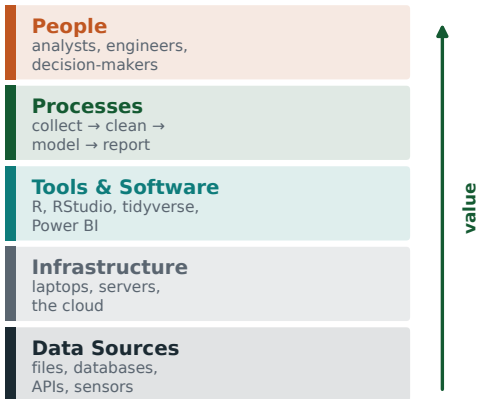
Most people arrive expecting data science to be modelling and machine learning.

Most of the job is getting messy data into a table that obeys the three rules.

Lectures 3 to 6 are devoted entirely to this.

The good news is that once the data is tidy, the analysis is often only a few lines.

3. The data science ecosystem



Everything and everyone it takes to turn raw data into a decision.

Notice that **tools are only one layer**. Clean data and a clear question will take you a lot further than expensive hardware.

Where this course sits

We work mostly in the **Tools** and **Processes** layers: R, RStudio, and the six-step workflow you just saw.

4. What is R?

R

A **programming language** built for statistics, data analysis and graphics, together with the software that runs it.

Why we use it in this course

- **Free and open-source**, yours for life, no licence
- Runs on **Windows, macOS and Linux**
- Built *for* statistics, not adapted to it
- Thousands of add-on **packages**
- A standard in research and in a great many companies

CRAN (Comprehensive R Archive Network) | the official site that stores R and its packages, where you will download from today

Download R: <https://cloud.r-project.org>

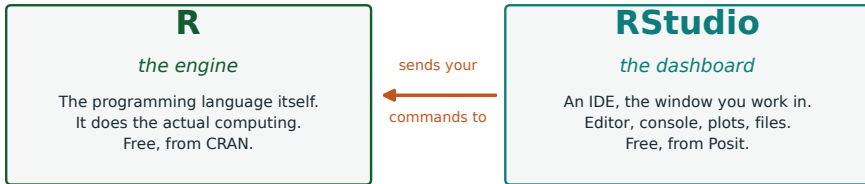


Jargon, unpacked

GUI (Graphical User Interface), anything you operate by clicking buttons, menus and windows rather than typing commands.

Plain R comes with a very basic GUI. It works, but it is bare, which is why almost nobody uses it directly.

4. R vs. RStudio, you need both



You install BOTH. R first, then RStudio.

A car without an engine does not move, and RStudio without R does nothing.

IDE (Integrated Development Environment) | one program that gathers everything you need to write code: editor, run button, file browser, help and graphs, all in a single window

RStudio is made by **Posit**.
It is also free.

4. The RStudio window

Source / Editor

Where you WRITE and SAVE
your script (.R file).
Code here is kept.

Environment / History

Every object you have created.
Your data lives here.

Console

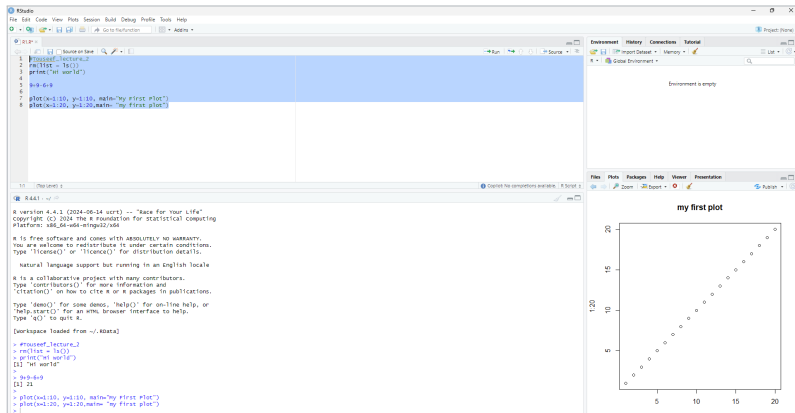
Where code RUNS.
Type here for quick tests,
nothing is saved.

Files / Plots / Packages / Help

Your folders, your graphs,
installed packages,
and documentation.

Type experiments in the **Console**. Keep anything worth repeating in the **Source** editor, and save it.

4. The RStudio window, the real thing



By the end of today your screen should look about like this.

Top left

Source editor, the saved script.

Bottom left

Console, where the code ran.
You can see `[1] "Hi world"`
and `[1] 21`.

Top right

Environment, empty here
because nothing has been
stored yet.

Bottom right

Plots, showing the graph the
last command drew.

Data Lab

Install R and RStudio on your own laptop

Laptops open. We do this together.

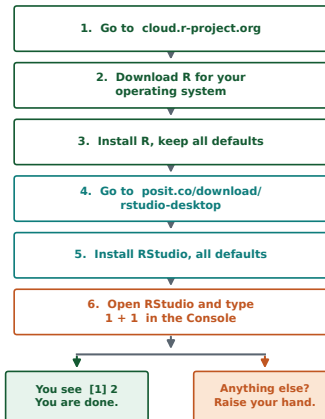
Data Lab: the goal

You leave this room with

- ✓ R installed
- ✓ RStudio installed
- ✓ both of them opening without an error
- ✓ your first line of R code run

If you get stuck

Just raise your hand. Please do not sit quietly and fall behind, because everything in Lecture 3 builds straight on this.



Install R BEFORE RStudio, the order matters.

Lab step 1: Install R

Go to <https://cloud.r-project.org> and pick your operating system.



The screenshot shows the R Project website. On the left is a sidebar with the R logo and various links. The main content area is titled 'The Comprehensive R Archive Network' and contains two sections: 'Download and Install R' and 'Source Code for all Platforms'.

CRAN
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About R
[R Homepage](#)
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Software
[R Sources](#)
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[Packages](#)
[Task Views](#)
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Documentation
[Manuals](#)
[FAQs](#)

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux \(Debian, Fedora/Redhat, Ubuntu\)](#)
- [Download R for macOS](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

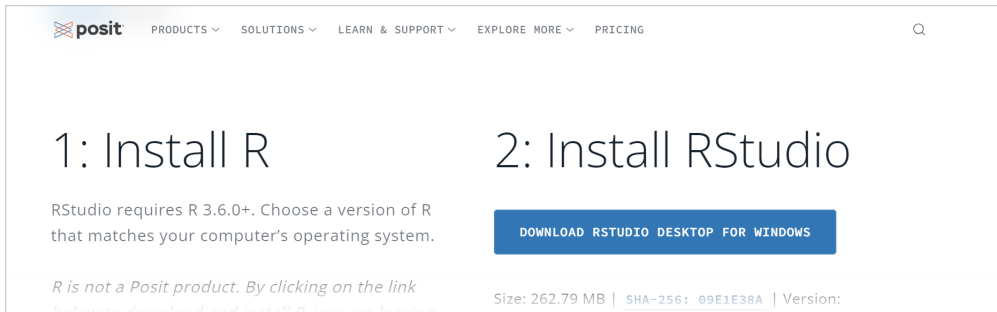
Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2024-06-14, Race for Your Life) [R-4.4.1.tar.gz](#), read [what's new](#) in the latest version.
- The CRAN directory [src/base-prerelease](#) contains R alpha, beta, and rc releases as daily snapshots in time periods before a planned release.

Accept every default in the installer. Do **not** change the installation folder.

Lab step 2: Install RStudio

Go to <https://posit.co/download/rstudio-desktop/> and download RStudio Desktop.



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1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are installing

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 262.79 MB | SHA-256: 09E1E38A | Version:

The page detects your system automatically. **Install R first**, RStudio cannot find an engine that is not there yet.

Lab step 3: Check that it works

Open RStudio. Click into the **Console** pane (bottom left) and type:

```
1 + 1
```

```
[1] 2
```

If you see that, your installation is complete.

✓ **You are ready for Lecture 3.**

What is that [1]?

R is telling you that this line of output begins at the first element of the result. Ignore it for now, it will make sense once results get longer.

Common install problems

RStudio opens but nothing runs, R itself is not installed. Go back to step 1.

Console shows + and will not run, R is waiting for you to finish the command. Press **Esc**.

What we covered today

Concepts

- ✓ variable, observation, value, dataset, database
- ✓ the data science process flow
- ✓ the three rules of tidy data
- ✓ wrangling = tidy + transform
- ✓ the data science ecosystem

Tools

- ✓ R, the language (the engine)
- ✓ RStudio, the IDE (the dashboard)
- ✓ CRAN, where R comes from
- ✓ the four panes of RStudio
- ✓ both installed and running

If your laptop shows `[1] 2` in the console, today worked.

Coming up: the next two lectures

Lecture 3 | Wed 16 Sept | Room G102

Introduction to the Tidyverse

- what a *package* is, and how to install one
- the `tidyverse`, the family of packages this course is built on
- *tibbles*, the modern data table
- **Data Lab:** installing the tidyverse

Lecture 4 | Fri 18 Sept | Room S217

Importing and transforming data with `dplyr`

- reading a real `.csv` file into R
- the verbs that do the work: `filter()`, `select()`, `mutate()`, `arrange()`, `group_by()`, `summarise()`
- chaining them together with the pipe

Before you come back

1. Finish the installation if you did not get through it today.
2. Open RStudio at home and check `1 + 1` still works.
3. Bring your laptop, charged.

The thread that connects them

Today you learned *what* tidy data is.

The next two lectures are about *how* to make messy data tidy, the 50–80% of the job.

Reading: *R for Data Science* (2e), Ch. 2, free at <https://r4ds.hadley.nz>

Sources used in this lecture

Course reading

- Wickham, H., Çetinkaya-Rundel, M. & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly. Introduction (the process flow) | Ch. 5 *Data Tidying* (the three rules). Free online: <https://r4ds.hadley.nz>
- Wright, C., Ellis, S., Hicks, S. & Peng, R. (2021). *Tidyverse Skills for Data Science in R*. <https://jhudatascience.org/tidyversecourse/>

Software, download links

- R, from CRAN: <https://cloud.r-project.org>
- RStudio Desktop, from Posit: <https://posit.co/download/rstudio-desktop/>

Worth bookmarking

- Wickham, H. (2014). Tidy Data. *Journal of Statistical Software*, 59(10). <https://www.jstatsoft.org/article/view/v059i10>
- Illustrations by Allison Horst: <https://allisonhorst.com>

Thank you

See you on 16 September for Lecture 3

Touseef Hameed | touseef.hameed@inn.no